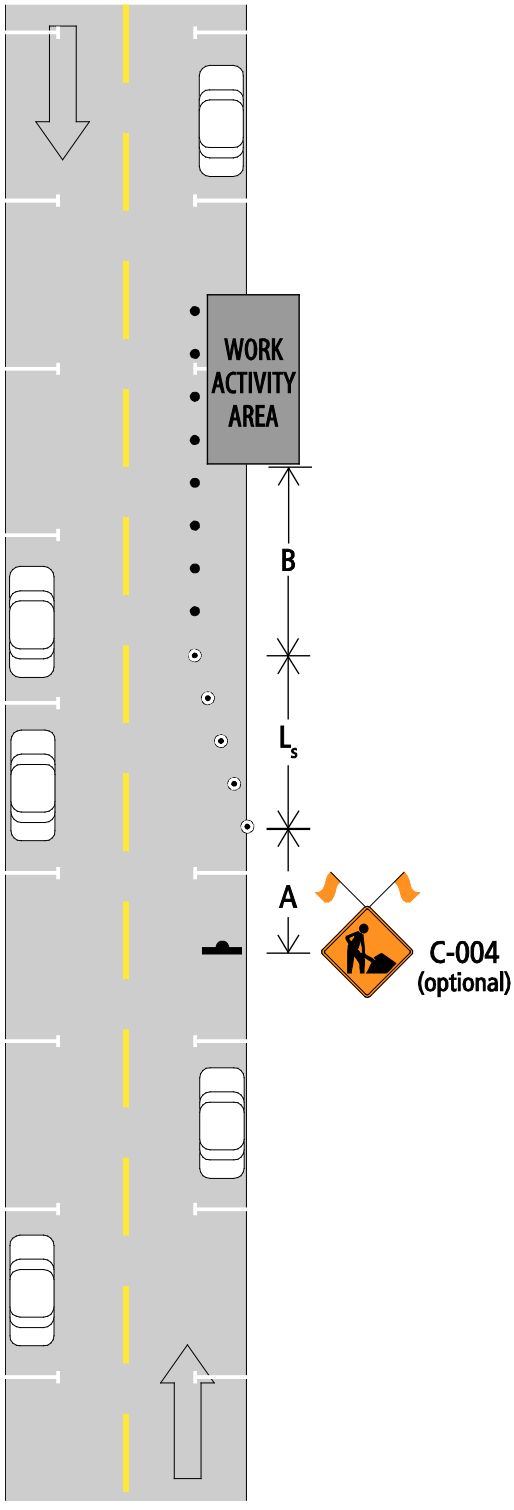
Figure 7.6: Work in Parking Lane – Urban Area



7.7 Roadside Work – Encroachment into Travel Lane – Short Duration

Purpose:

This layout shows a typical setup on a low-speed (≤ 60 km/h), low-volume roadway where work encroaches into the travelled portion of the roadway but sufficient space remains for vehicles to pass the works within their own lanes.

A low-volume roadway is one on which the total roadway volume is <1,000 vehicles per day. Traffic volumes may be obtained from the local Road Authority.

For higher-speed roadways, the options are to create a lowered Construction Speed Zone or to implement a lane closure.

Standard:

- At least 3 metres of width shall be maintained for each travel lane.
- Single lane alternating setup shall be used if at least 3 metres of width cannot be maintained for each travel lane.

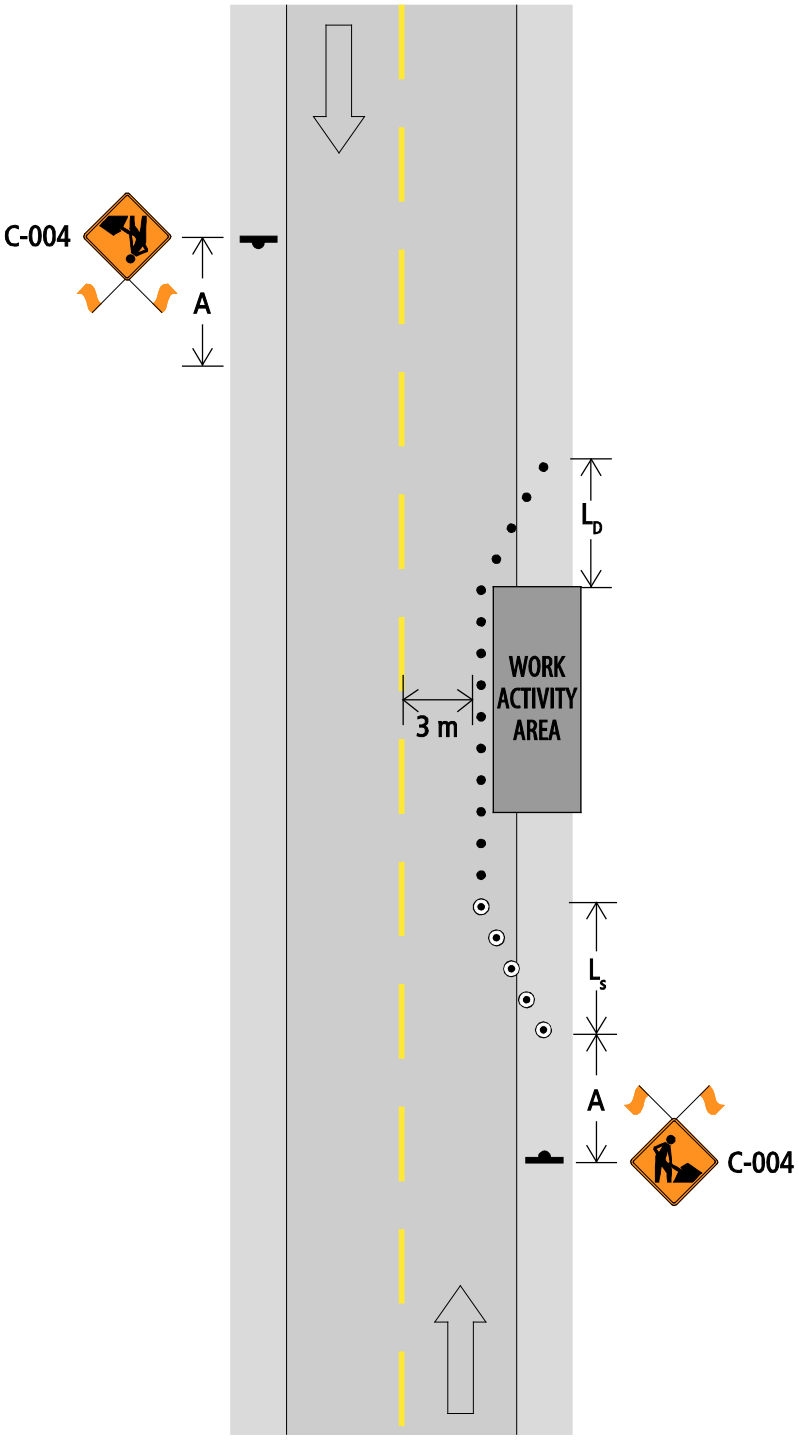
Guidance:

- None at this time.

Options:

- Additional advance warning may be appropriate, such as a Road Narrows C-134 sign between the Crew Working Ahead C-004 sign and the taper.
- The taper and channelizing devices may be omitted if a shadow vehicle with flashing arrow board (FAB) in caution mode is used.
- If the opposite shoulder is suitable for carrying vehicular traffic and at least 3 metres of width can be maintained for each travel lane, the centreline may be shifted by using closely-spaced channelizing devices.

Figure 7.7: Roadside Work – Encroachment into Travel Lane – Short Duration



7.8 Lane Closure with TCPs – Single Lane Alternating – Short and Long Duration

Purpose:

This layout shows the appropriate positions of TCPs when they are directing traffic for a lane closure on a two-lane, two-way roadway.

Standard:

- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.
- When used at night, the TCP station shall be illuminated with overhead lighting.
- Barricades are required at each end of the work activity area for long-duration work.

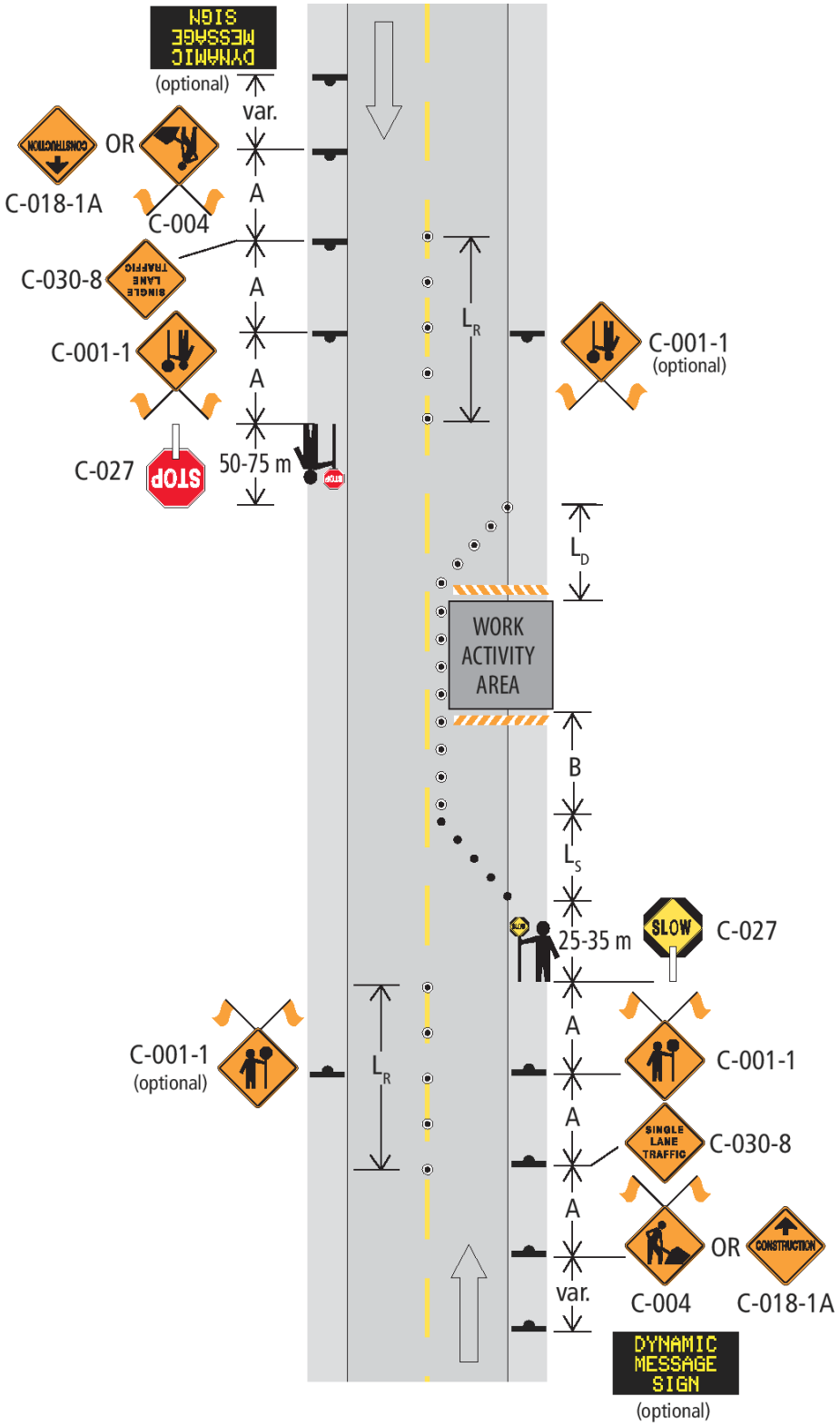
Guidance:

- The distance between the TCP and the Traffic Control Person Ahead C-001-1 sign should not exceed 150 metres.
- Where Crew Working – Maximum Speed C-002-2 signs establish a Temporary Speed Zone, the C-002-2 should be placed upstream of the C-004 or C-018-1A.
  - Thank You Resume Speed C-086-1 signs should be placed across from the Crew Working – Maximum Speed C-002-2 signs in the opposing lanes.

Options:

- An additional Traffic Control Person Ahead C-001-1 sign may be added to the far side of the road to provide queued drivers with increased awareness of the TCP position.
- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- A Prepare to Stop C-029 sign may replace the Single Lane Traffic C-030-8 sign for other applications that require traffic to stop (e.g., equipment crossing road).
- Run-in delineation, L_R may be omitted during period where queues are low, permitting the full release of the queue in each direction.

Figure 7.8: Lane Closure with TCPs – Single Lane Alternating – Short and Long Duration



7.8.1 Lane Closure with TCPs – Single Lane Alternating with Speed Reduction (Construction Speed Limit ≤ 50 km/h)

Purpose:

This layout shows the signs used and the appropriate positions of TCPs when they are directing traffic for a lane closure on a two-lane, two-way roadway. This layout should be used when there is a speed reduction and the construction speed limit is ≤ 50 km/h.

Construction speed limits are regulatory speeds established in Construction Speed Zones within long-term construction and maintenance project areas where there are continuous hazards for motorists or where workers are in close proximity to active travel lanes.

Construction Speed Zones should be applied prudently because **overuse reduces effectiveness.** Drivers should be able to perceive the need to reduce speed.

Standard:

- Conflicting speed limit signs within the Construction Speed Zone shall be covered or removed.
- When used at night, the TCP station shall be illuminated with overhead lighting.
- Barricades are required at each end of the work activity area for long-duration work.

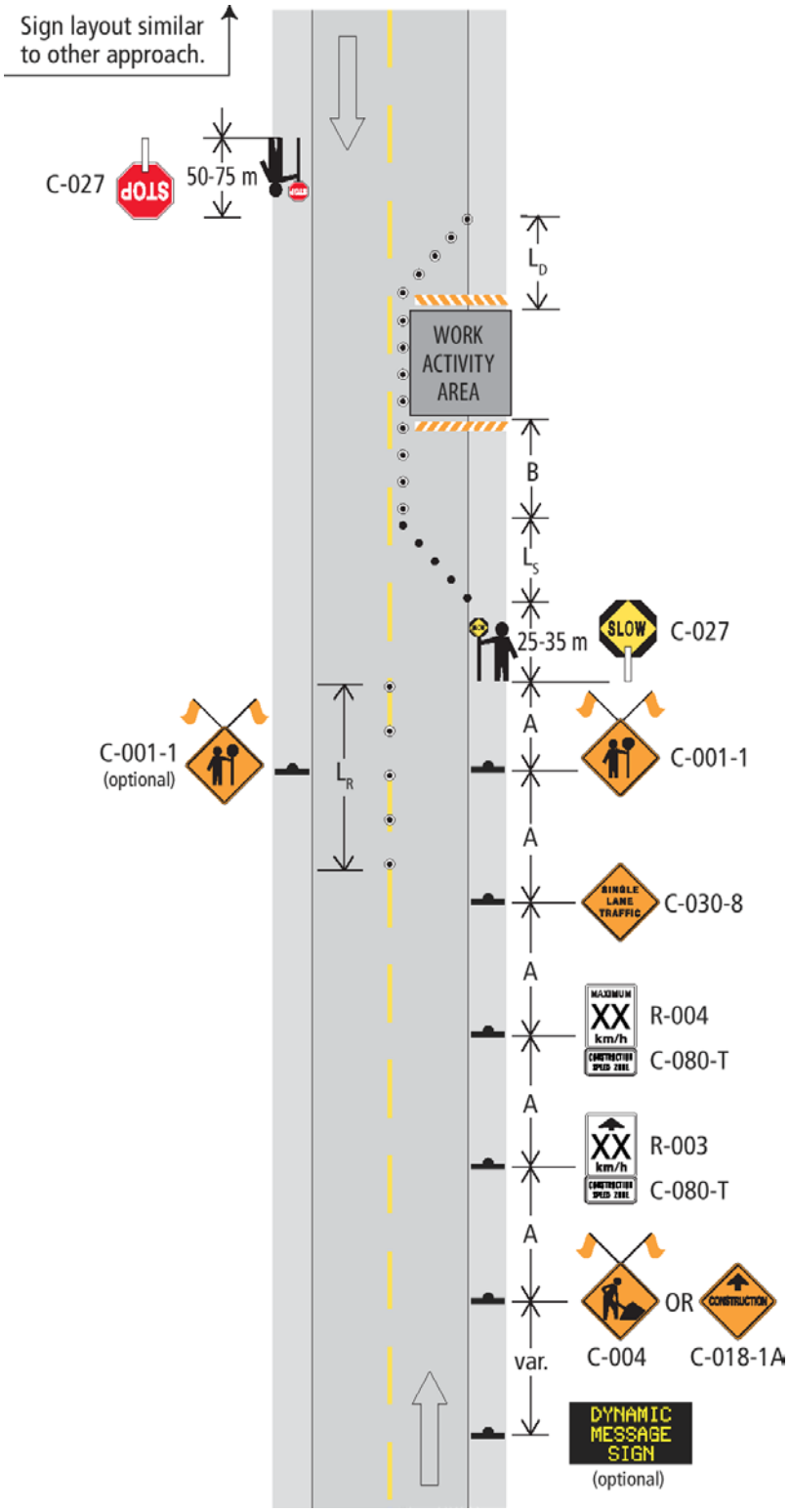
Guidance:

- The distance between the TCP and the Traffic Control Person Ahead C-001-1 sign should not exceed 150 metres.
- Construction speed limit signs should be covered or removed when no work is occurring and other hazards are not present.
- It is important to record when construction speed limits are installed and covered/removed.
- Where Crew Working – Maximum Speed C-002-2 signs establish a Temporary Speed Zone, the C-002-2 should be placed upstream of the C-004 or C-018-1A.
  - Thank You Resume Speed C-086-1 signs should be placed across from the Crew Working – Maximum Speed C-002-2 signs in the opposing lanes.

Options:

- An additional Traffic Control Person Ahead C-001-1 sign may be added to the far side of the road to provide queued drivers with increased awareness of the TCP position.
- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- A Prepare to Stop C-029 sign may replace the Single Lane Traffic C-030-8 sign for other applications that require traffic to stop (e.g., equipment crossing road).
- If secondary signs are applied, they should be installed on the left side of the highway.
- Run-in delineation, LR may be omitted during period where queues are low, permitting the full release of the queue in each direction.

Figure 7.8.1: Lane Closure with TCPs – Single Lane Alternating with Speed Reduction (Construction Speed Limit ≤ 50 km/h)



7.8.2 Lane Closure with TCPs – Single Lane Alternating with Speed Reduction (Construction Speed Limit ≥ 60 km/h)

Purpose:

This layout shows the signs used and the appropriate positions of TCPs when they are directing traffic for a lane closure on a two-lane, two-way roadway. This layout should be used when there is a speed reduction and the construction speed limit is ≥ 60 km/h.

Construction speed limits are regulatory speeds established in Construction Speed Zones within long-term construction and maintenance project areas where there are continuous hazards for motorists or where workers are in close proximity to active travel lanes.

Construction Speed Zones should be applied prudently because **overuse reduces effectiveness.** Drivers should be able to perceive the need to reduce speed.

Standard:

- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.
- Conflicting speed limit signs within the Construction Speed Zone shall be covered or removed.
- When used at night, the TCP station shall be illuminated with overhead lighting.
- Barricades are required at each end of the work activity area for long-duration work.

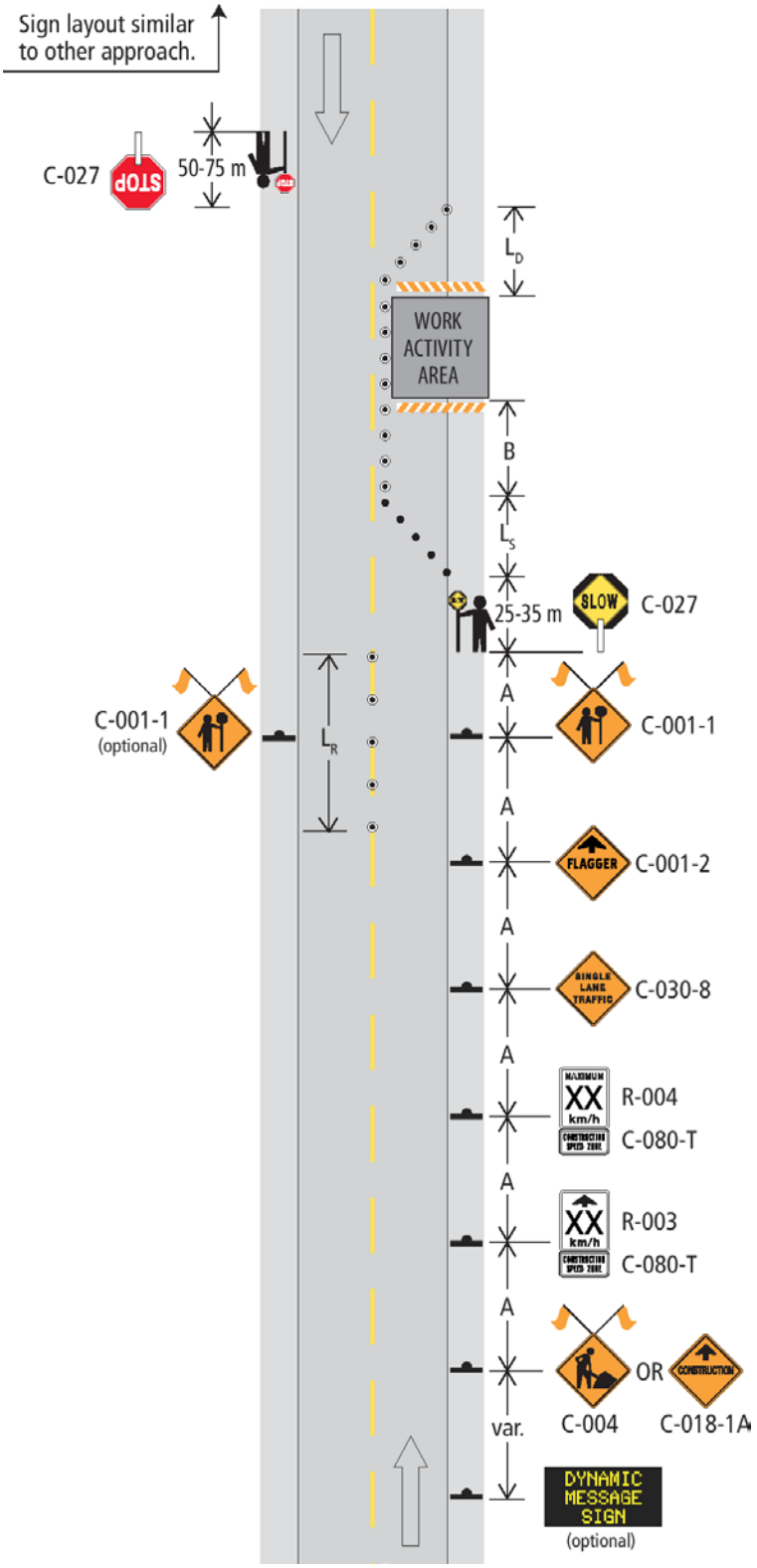
Guidance:

- The distance between the TCP and the Traffic Control Person Ahead C-001-1 sign should not exceed 150 metres.
- Construction speed limit signs should be covered or removed when no work is occurring and other hazards are not present.
- It is important to record when construction speed limits are installed and covered/removed.
- Where Crew Working – Maximum Speed C-002-2 signs establish a Temporary Speed Zone, the C-002-2 should be placed upstream of the C-004 or C-018-1A.
    - Thank You Resume Speed C-086-1 signs should be placed across from the Crew Working – Maximum Speed C-002-2 signs in the opposing lanes.

Options:

- An additional Traffic Control Person Ahead C-001-1 sign may be added to the far side of the road to provide queued drivers with increased awareness of the TCP position.
- The Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- A Prepare to Stop C-029 sign may replace the Single Lane Traffic C-030-8 sign for other applications that require traffic to stop (e.g., equipment crossing road).
- If secondary signs are applied, they should be installed on the left side of the highway.
- Run-in delineation, LR may be omitted during period where queues are low, permitting the full release of the queue in each direction.

Figure 7.8.2: Lane Closure with TCPs – Single Lane Alternating with Speed Reduction (Construction Speed Limit ≥ 60 km/h)



7.9 Lane Closure with AFADs – Short and Long Duration

Purpose:

This layout shows the use of Automated Flagger Assistance Devices (AFADs) when they are used to control traffic for a lane closure on a two-lane, two-way roadway.

Standard:

- When used at night, the AFAD station shall be illuminated with overhead lighting.
- A black-on-white STOP HERE ON RED or STOP HERE ON RED SIGNAL sign shall be installed on the right side of the approach at the point where drivers are expected to stop. It may be installed on the AFAD device itself.
- When AFADs are used, the construction speed limit shall be ≤ 70 km/h.

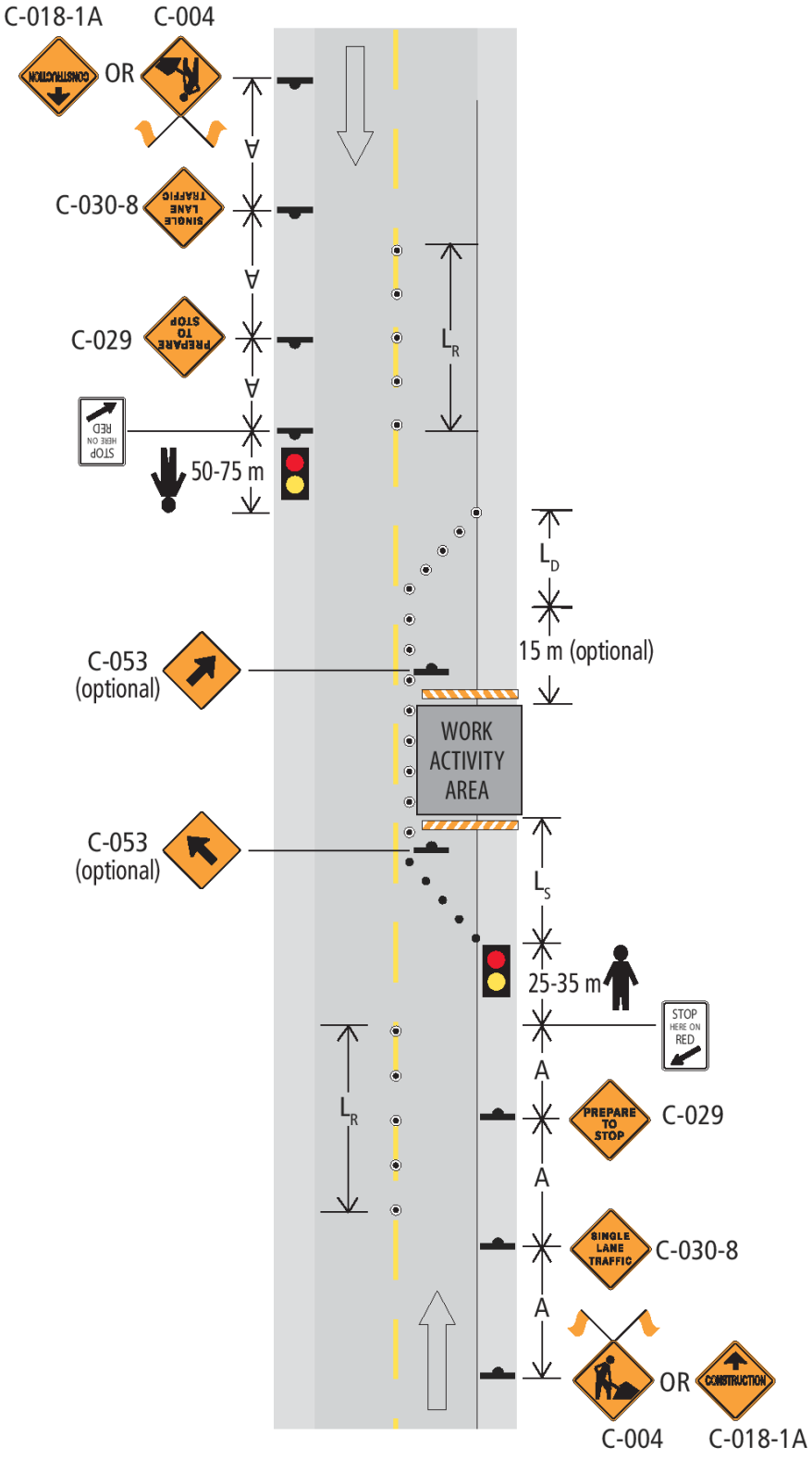
Guidance:

- Where Crew Working – Maximum Speed C-002-2 signs establish a Temporary Speed Zone, the C-002-2 should be placed upstream of the C-004 or C-018-1:
  - Thank You Resume Speed C-086-1 signs should be placed across from the Crew Working – Maximum Speed C-002-2 signs in the opposing lanes.

Options:

- A Prepare to Stop C-029 sign may replace the Single Lane Traffic C-030-8 for applications other than single lane alternating traffic where traffic is required to stop (e.g., equipment crossing road).
- Lane Closure Arrow C-053 signs may be added upstream and downstream of the work activity area to direct traffic into the correct lanes.
- Run-in delineation, L_R may be omitted during period where queues are low, permitting the full release of the queue in each direction.

Figure 7.9: Lane Closure with AFADs – Short and Long Duration



7.10 Lane Closure with Temporary Signals – Single Lane Alternating – Short and Long Duration

Purpose:

This layout shows the appropriate placement of temporary traffic signals when they are required to control traffic for a lane closure on a two-lane, two-way roadway.

Standard:

- Temporary traffic control signals shall be installed and operated in accordance with *Section 4.8: Portable Traffic Signals.*
- Signal timing and signal head locations shall be established by qualified personnel.
- Advance warning shall include a Signal Ahead C-112 sign and a Single Lane Traffic C-030-8 sign.
- An overhead advance warning W-012 sign, with flashers, shall be used in advance of the stop bar in speed zones ≥ 70 km/h or as required by the Road Authority.
- A Stop Line Here R-025-R sign is required to advise drivers of where to stop.
- Barricades are required at each end of the work activity area for long-duration work.

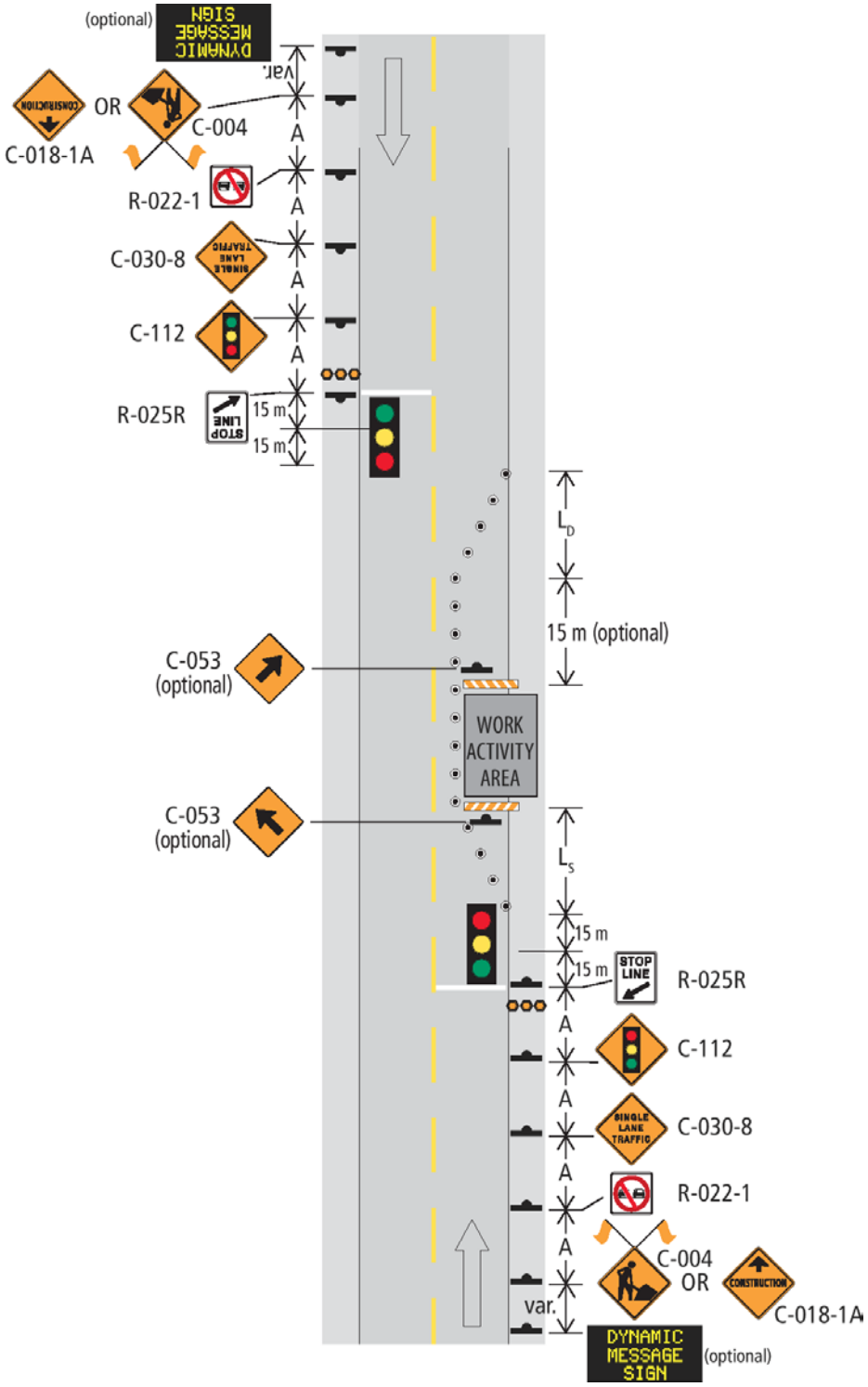
Guidance:

- Inspection of the portable traffic signal (PTS) should occur at least once daily and should include, at minimum, checking the traffic operation (vehicle delay and throughput), signal alignment, power supply, and evidence of vandalism.
- Flexible drum delineators should be placed on the approach side of the portable traffic signal (PTS) to provide notification and protection for road users, including cyclists.

Options:

- Lane Closure Arrow C-053 signs may be added upstream and downstream of the work activity area to direct traffic into the correct lanes.
- A Passing Permitted R-023 sign may be used when traffic exits the work zone.
- A stop bar may be installed to accompany the Stop Line Here R-025-R sign when possible, especially for long duration work taking place over several days. Removable pavement markings may be used to establish the stop bar.

Figure 7.10: Lane Closure with Temporary Signals – Single Lane Alternating – Short and Long Duration



7.11 Work on Low-Volume Roadway – No Centreline – Short Duration

Purpose:

This layout shows a typical setup on a low-volume roadway where work encroaches into the travelled portion of the roadway but sufficient space remains for vehicles to pass the works within their own lanes.

A low-volume roadway is one on which the total roadway volume is <1,000 vehicles per day. Traffic volumes may be obtained from the local Road Authority.

Standard:

- A 5.5 m minimum roadway width shall be maintained.

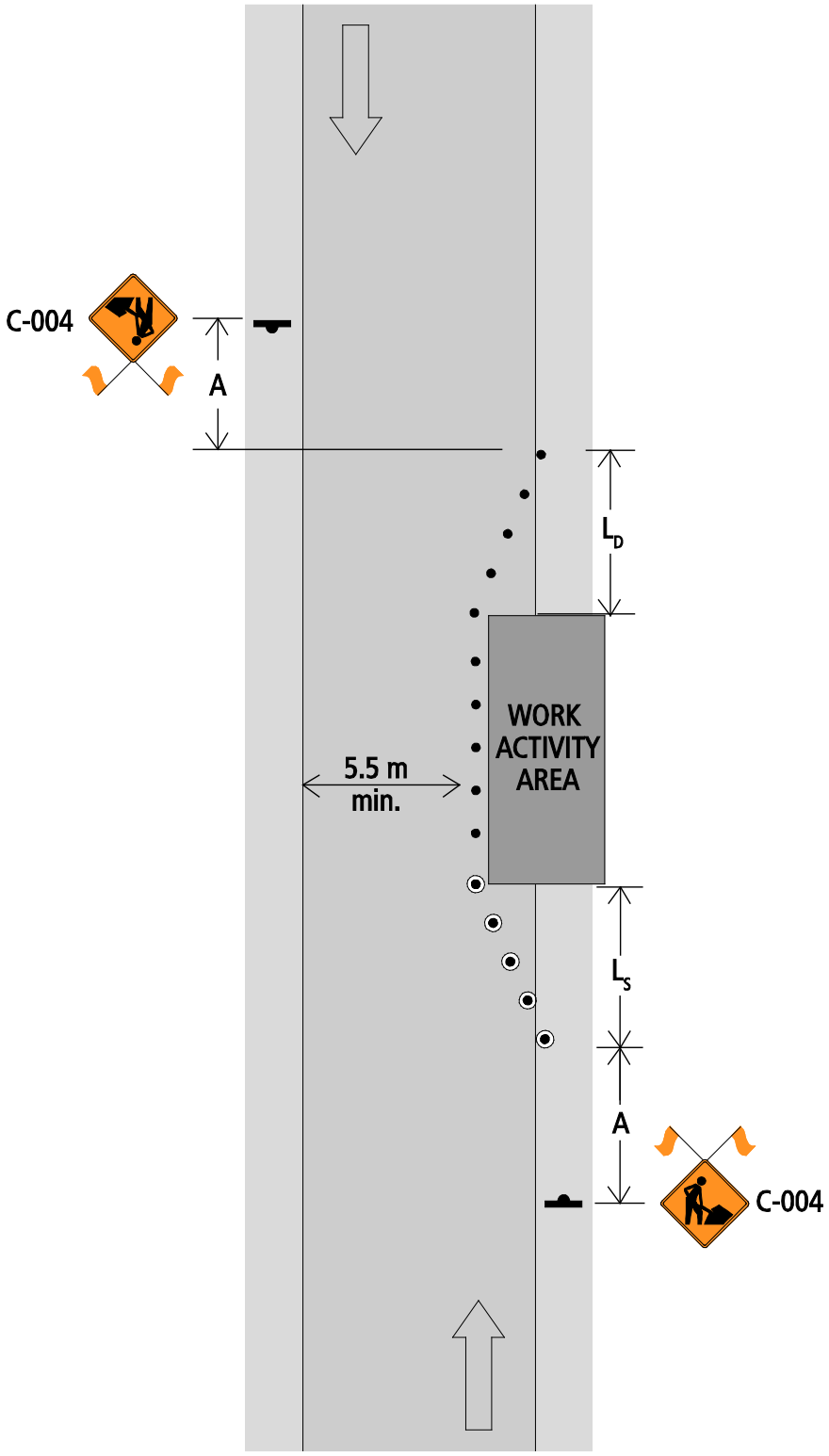
Guidance:

- Any stopped work vehicle should keep as far right as practicable by using shoulder space whenever possible.
- For speeds ≥ 70 km/h, a buffer space should be used.

Options:

- For speeds ≤ 60 km/h, tubular markers may be used instead of drums for leading tapers, and cones may be used instead of tubular markers for other channelizing devices.
- A vehicle-mounted dynamic message sign (DMS) or a flashing arrow board (FAB) in caution mode may be positioned within the taper.

Figure 7.11: Work on Low-Volume Roadway – No Centreline – Short Duration



7.12 Work on Low-Volume Roadway – No Centreline – Long Duration

Purpose:

This layout shows a typical setup on a low-volume roadway where work encroaches into the travelled portion of the roadway but sufficient space remains for vehicles to pass the works within their own lanes.

A low-volume roadway is one on which the total roadway volume is <1,000 vehicles per day. Traffic volumes may be obtained from the local Road Authority.

Standard:

- A Road Narrows Ahead C-134 sign is required for both directions of travel.
- A Flashing Arrow Board (FAB) is required if workers are present.
- Barricades are required at each end of the work activity area for long-duration work.

Guidance:

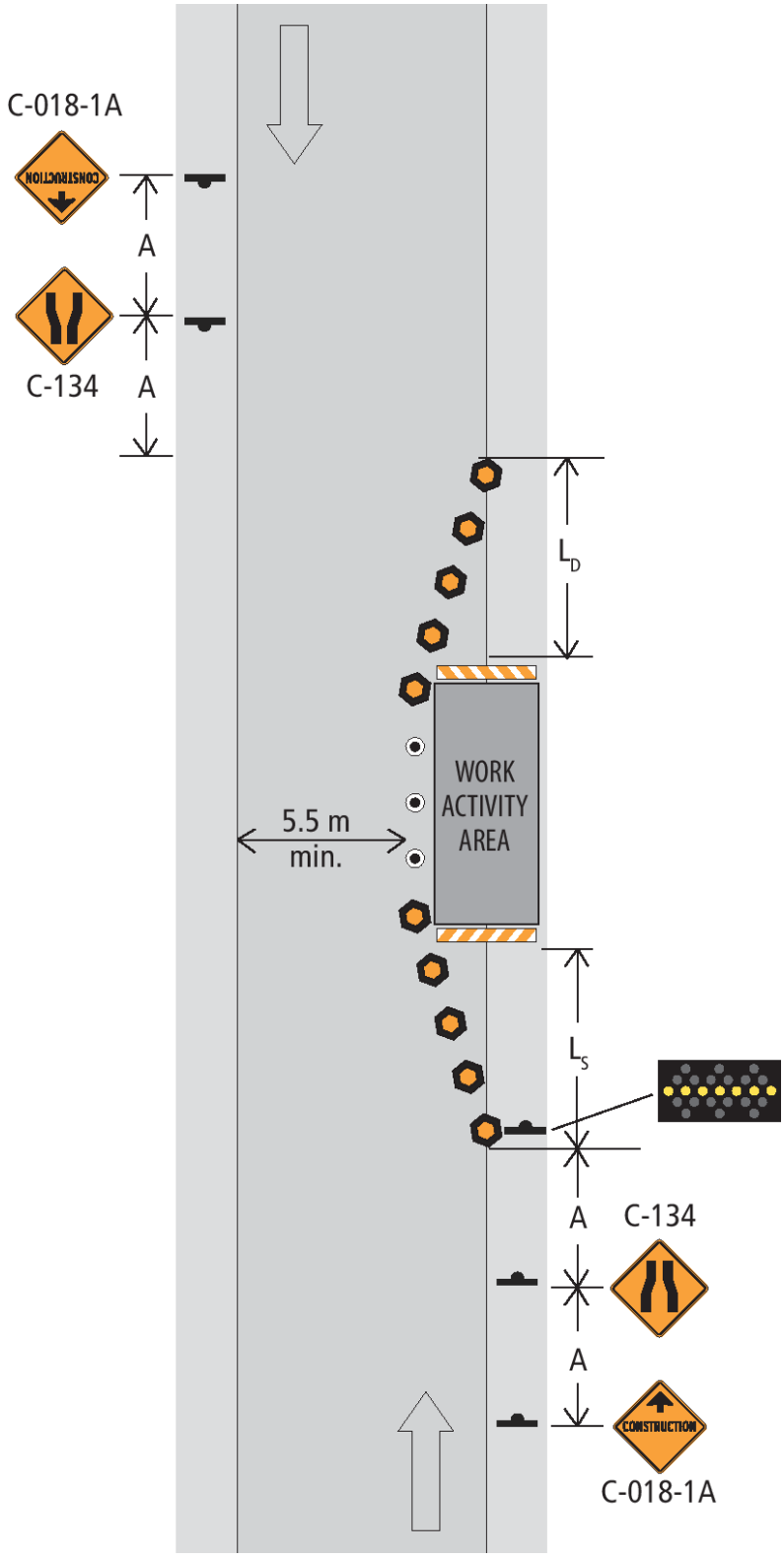
- If the opposite shoulder is suitable for carrying vehicles and at least 3.5 metres of space can be maintained for each travel lane, a lane shift may be implemented using the appropriate Lane Shift C-117-L/R signs.
- Any stopped work vehicle should keep as far right as practicable by using shoulder space whenever possible.

Options:

- The flashing arrow board (FAB) may be replaced as shown below.

	Workers or Work Vehicles Present	No Workers or Vehicles Present
Day	360^0 flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign.
Night	Use flashing arrow board (FAB).	Type A yellow warning lights on barricades and drums.

Figure 7.12: Work on Low-Volume Roadway – No Centreline – Long Duration



7.13 Two-Way Left-Turn Lane Closed – Short and Long Duration

Purpose:

This layout shows the typical setup for the closure of a two-way left-turn lane. If the work will encroach into either travel lane, a left lane closure for the affected lane(s) should be considered.

Standard:

- A flashing arrow board (FAB) in caution mode is required for each direction of travel.
- Barricades are required at each end of the work activity area for long-duration work.

Guidance:

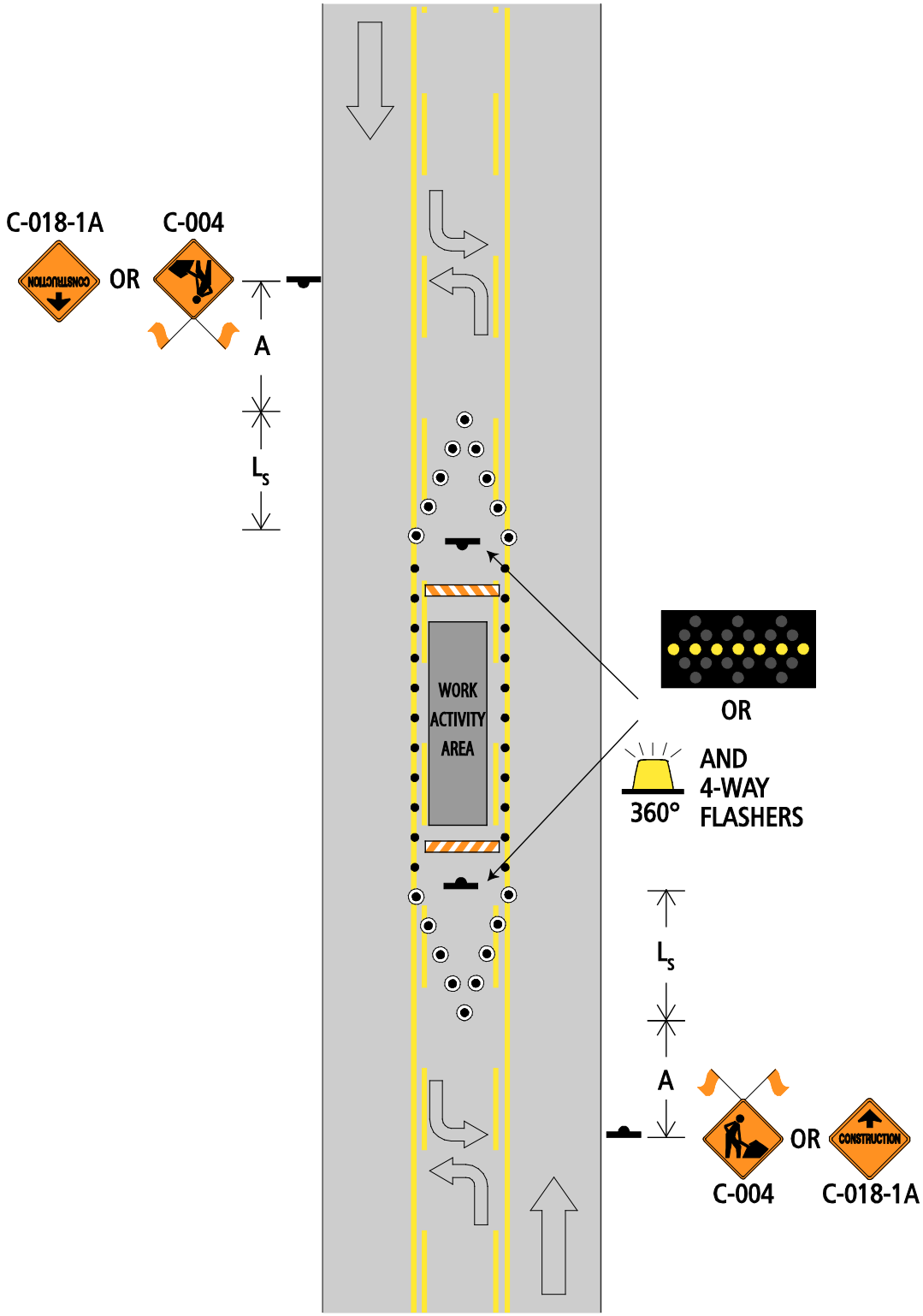
- Left-turning movements should be prohibited along the work activity area and taper.

Options:

- Depending on the nature of the work, one or both adjacent lanes may also have to be closed.
- For speeds ≤ 60 km/h, cones instead of tubular markers may be used to channelize traffic.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning lights.
≥ 70 km/h	No substitution.	

Figure 7.13: Two-Way Left-Turn Lane Closed – Short and Long Duration



7.14 Roadside Diversion – Long Duration

Purpose:

A roadside diversion provides a new alignment around the work area, typically adjacent to the original alignment.

Standard:

- A Road Diversion Ahead C-052-L/R sign shall be used in advance of the diversion to note the direction of the new alignment.
- Pavement markings no longer applicable to the traffic pattern of the roadway shall be covered, removed, or eradicated.
- If the diversion is paved, temporary pavement markings are required to mark the centreline (see *Section 4.4: Pavement Markings*).
- If the diversion is gravel, a Pavement Ends C-149 sign is required in advance of the graveled portion.
- Type 3 barricades shall be used to mark the closed portion of the roadway, with a Road Closed C-030-6A sign and a Detour Right C-006-R marker mounted on the barricades.
- The edges of the diversion shall be defined using channelizing devices or barriers.

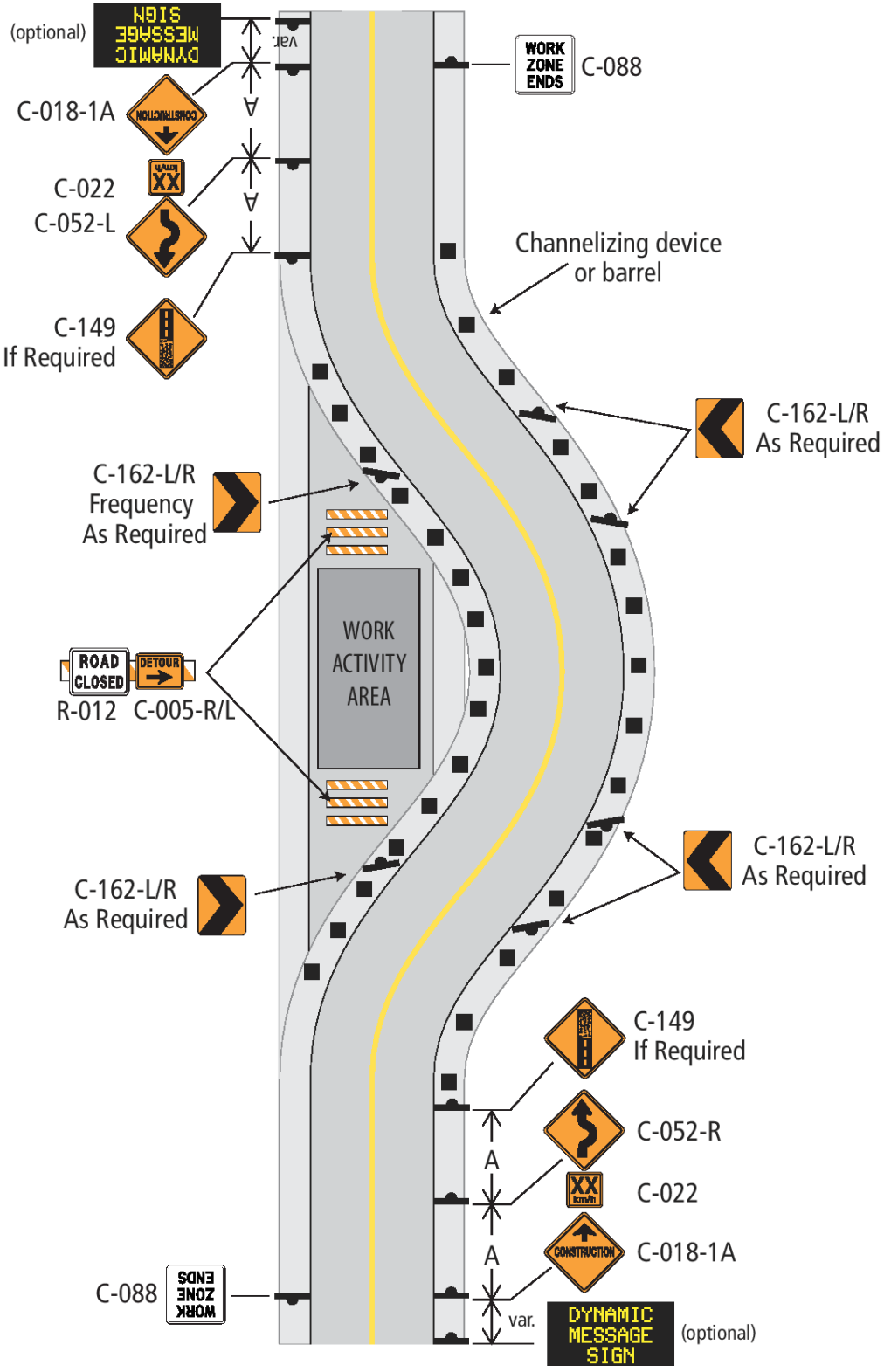
Guidance:

- A No Passing Zone should be created through the diversion by posting No Passing R-022-1 signs at the beginning of the diversion and periodically along the diversion route if required.
- If the tangent distance along the temporary diversion is more than 200 metres, chevrons should be applied separately for each curve.
- An Advisory Speed C-022 tab should be posted as recommended by a Traffic Engineer.

Options:

- Supplemental delineation devices such as additional chevrons, delineators, or raised pavement markers (RPMs) may be required.
- If the diversion is not paved, a centreline may be established by placing cones or tubular markers where a centreline would be.

Figure 7.14: Roadside Diversion – Long Duration



7.15 One-Lane Bridge or Roadway – Short and Long Duration

Purpose:

This layout shows the typical sign setup for traffic self-regulation as drivers approach a one-lane bridge or a one-lane section of roadway. It is typically used on low-volume roadways.

Standard:

- A Yield to Oncoming Traffic R-056-1 sign is used on the side of the bridge or the lane with the longest sight distance.
- A Narrow Structure Ahead C-135 sign shall be used for a one-lane bridge.
- A Road Narrows Ahead C-134 sign shall be used for a one-lane (narrow) road.
- On a roadway reduced to one lane, drums are required for the leading taper.
  - Flashing lights are required on taper devices for setups left in place overnight.
  - Where the speed limit is ≥ 70 km/h, a buffer space shall be used.
- Barricades are required at each end of the work activity area for long-duration work.

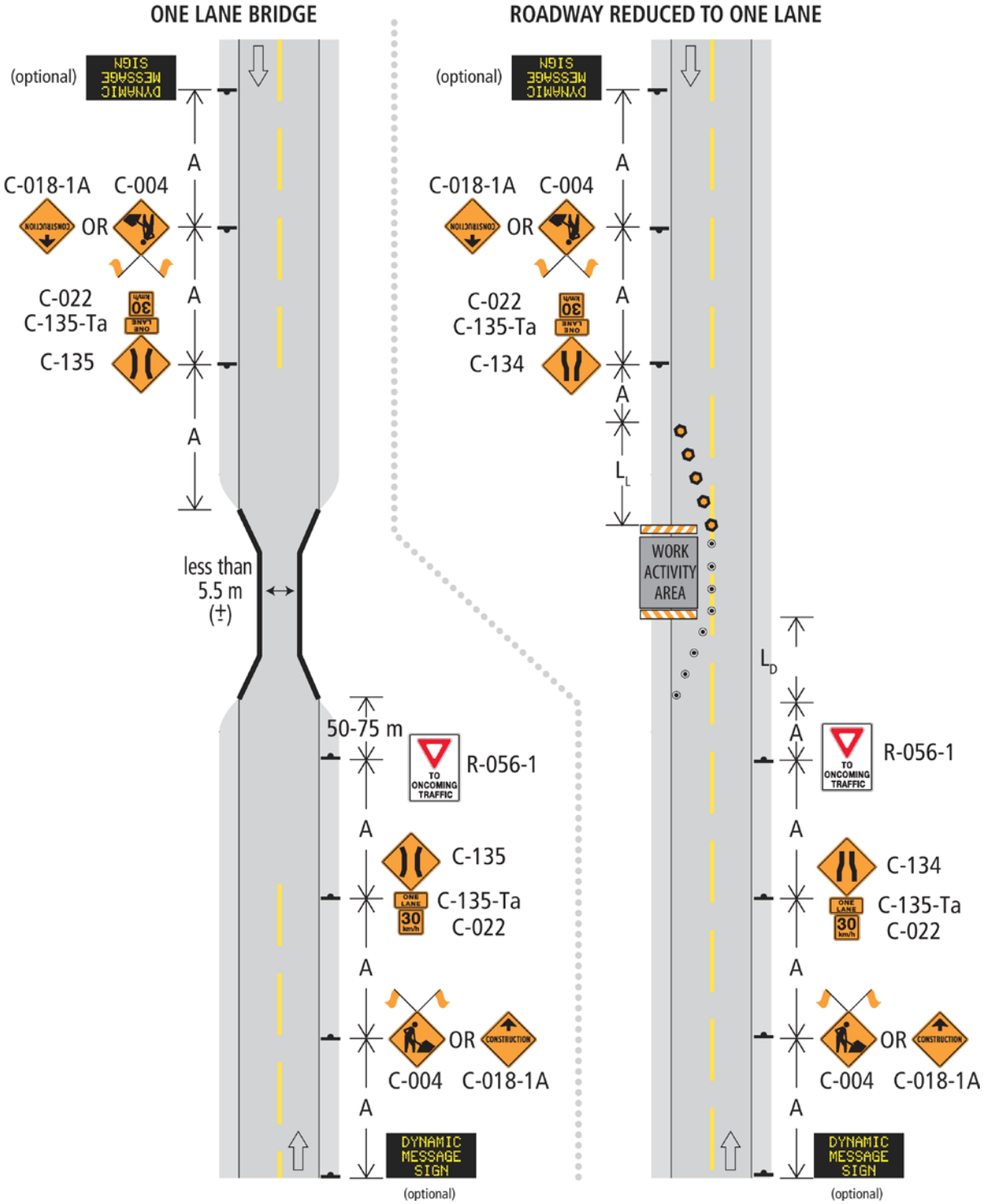
Guidance:

- Advisory speed tabs are typically posted at 30 km/h.
- Grade and curvature of the roadway should be considered to determine the appropriate direction of the R-056-1 sign.
- Where traffic volumes are higher or sight distance is limited causing difficulty navigating the work zone, other traffic control measures should be considered, such as portable traffic signals or TCPs.

Options:

- Additional advance warning may include a dynamic message sign (DMS), a Reduce Speed C-032 sign, or a Traffic Pattern Change C-063 sign.

Figure 7.15: One-Lane Bridge or Roadway – Short and Long Duration



7.16 Pilot Cars

Purpose:

A pilot car (pilot vehicle) is used to lead drivers through a work zone where traffic is single lane alternating where traffic volumes and conditions allow.

The work activity area is typically more than one kilometre long, or the complexity of the work activity area makes it difficult for drivers to navigate on their own (e.g., where there is a substantial change in alignment).

Using a pilot car may reduce the risk of incidents, help to prevent traffic from straying onto the work site, and help to manage driver compliance with construction speed limits.

See also *Section 4.11.9: Pilot Cars for Work Zones.*

Standard:

- The Follow Pilot Car C-049 sign shall be used at each departure point.
- Pilot cars shall have 360-degree flash light and 4-way flashers. The double-sided Pilot Car C-048-1-DS sign or the Pilot Car C-048-2 overhead sign is required on each pilot car.
- TCPs shall regulate traffic at each end of the work zone.
- TCPs and pilot car operators shall remain in communication throughout the work zone.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.

Guidance:

- Additional TCPs and Follow Pilot Car C-049 signs should be used between the TCPs at each end of the work zone for both of these situations:
  - at every intersection that may require additional guidance for motorists entering the roadway
  - at business driveways with sufficiently high traffic volumes, where there is a risk that motorists could enter the roadway against the flow of traffic
- Confirmatory Follow Pilot Car C-049 signs and intermittent centreline delineation should be used for long work zones to remind drivers of the pilot car operation.
- The travel speed should minimize gaps between the vehicles in the platoon to help prevent tar splatter, losing cars, or creating dust in the work zone.
  - When a pilot car is not operating, its 360-degree rotating lights and 4-way flashers should be turned off.
- Where cyclists are regularly observed using the shoulder, provisions for transporting cyclists past the work should be considered.

Options:

- To remind drivers that passing is not allowed within the zone, it may be beneficial to post No Passing R-022-1 signs through the site.
- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- A second Pilot Car may be used to follow or chase the back of the queue to maintain control and visibility of all vehicles.
- Temporary Traffic Signals may be used, controlled by the Pilot Car driver, to replace the TCP

Figure 7.16: Pilot Cars

